

Reliable and Economic Island, Microgrid and Grid Supply with Renewable Energy using Electrochemical Storage

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Competence E



Campus North

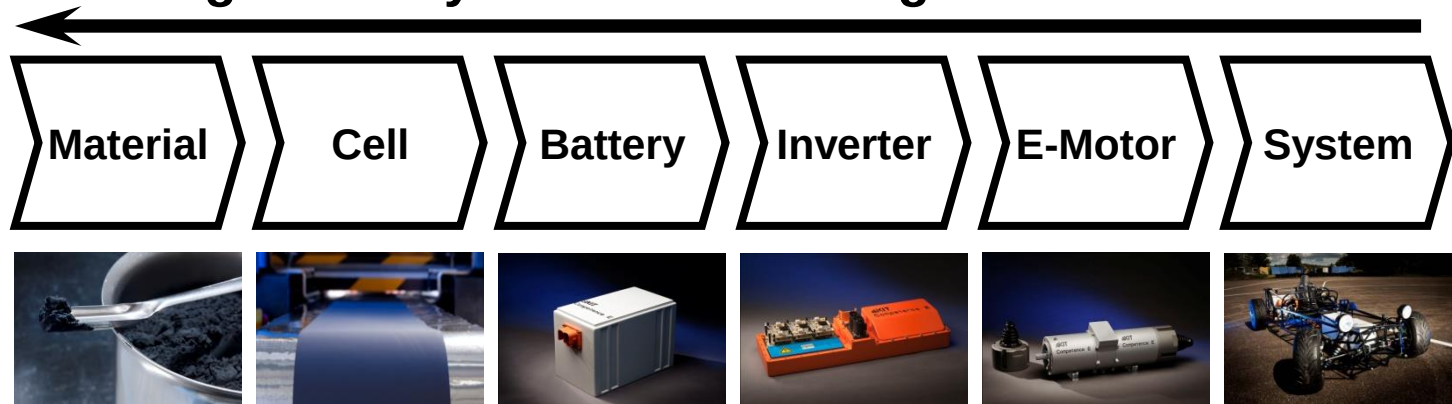


Campus South

- University AND Federal Gov. Lab
- 9.000 Employees
- 700 Mio.€ Budget
- 160 Institutes
- 372 Professors
- 22.800 Students

Competence E: System Competence along the Value Creation Chain

Alignment by continuous management structure



New Materials

New Cell Designs

New Battery Designs

New Manufacturing Methods



**250 €/kWh at 250 Wh/kg on
system level in 2018**

Energy Strategy of the German Federal Government

Competence E

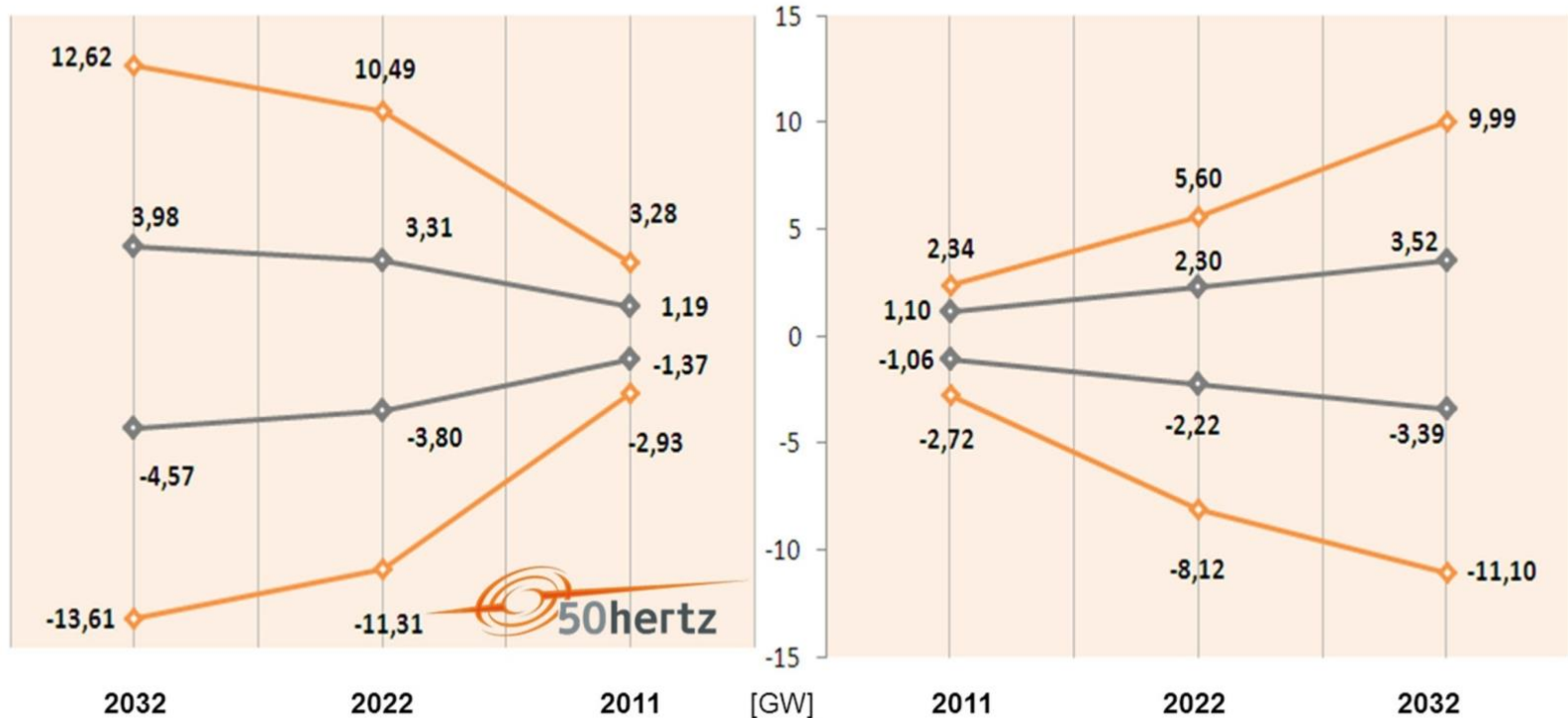
	GHG Emissions	Replacement of Fossiles by Renewables		Reduction of Fossiles Consumption				End of Nuclear
	Treibhausgas-Emissionen	Erneuerbare Energien		Minderung Energiebedarf				Kern-energie
		Brutto-Endenergie	Strom-erzeugung	Primär-energie	Gebäude-Wärme	End-energie Verkehr	Strom-verbrauch	
2011								-41%
2015								-47%
2017								-54%
2019								-60%
2020	-40%	18%	35%	-20%	-20%	-10%	-10%	
2021								-80%
2022								-100%
2030	-55%	30%	50%					
2040	-70%	45%	65%					
2050	-80 bis -95%	60%	80%	-50%	-80%	-40%	-25%	
Basis	1990	-	-	2008	2008	2005	2008	2010

Quelle: BReg 2010/2011, Berechnungen des Öko-Instituts Freiburg im Auftrag des BMU vom 28. März 2012

Maximum Power Ramps in German Electricity System

Maximum power ramps
caused by German PV power *

Maximum power ramps
caused by German wind power **



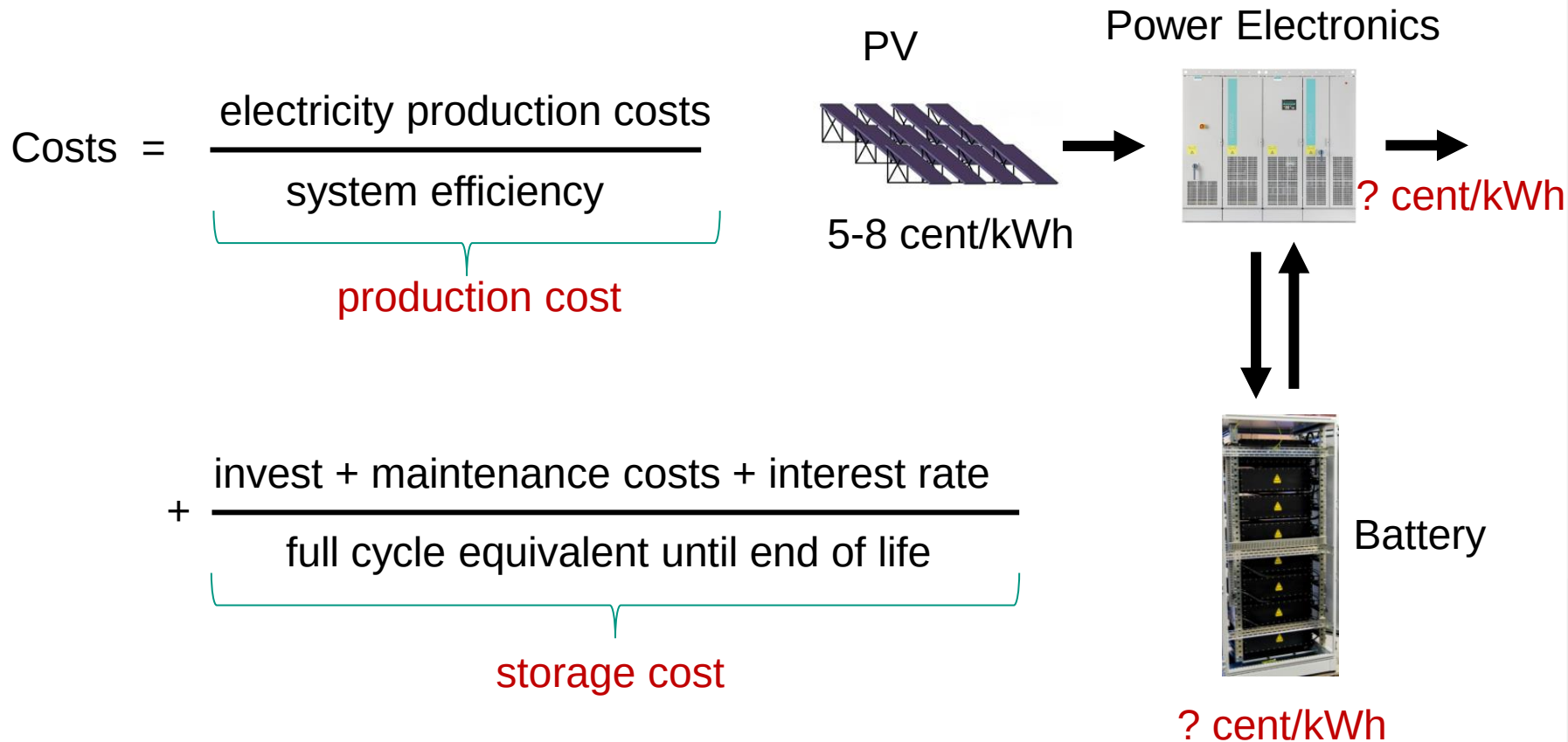
* Installed German PV power for 2011 was 19,47 GW (source: "Stammdatenbank" of the four TSO's)

** Installed German wind power for 2011 was 28,82 GW (source: IWES "Windenergiereport Deutschland 2011")

In 2022 and 2032 the data of the installed power as well as the 1 hour ramps can be found in "Leitszenario (B)" of the NEP (source: plan for the development of the German transmission grid: "Netzentwicklungsplan 2012"). The ¼ hour ramps for the years 2022 and 2032 are linearly extrapolated by using the maximal ramps per ¼ hour of 2011 (source: feed-in data of the four German TSO's).

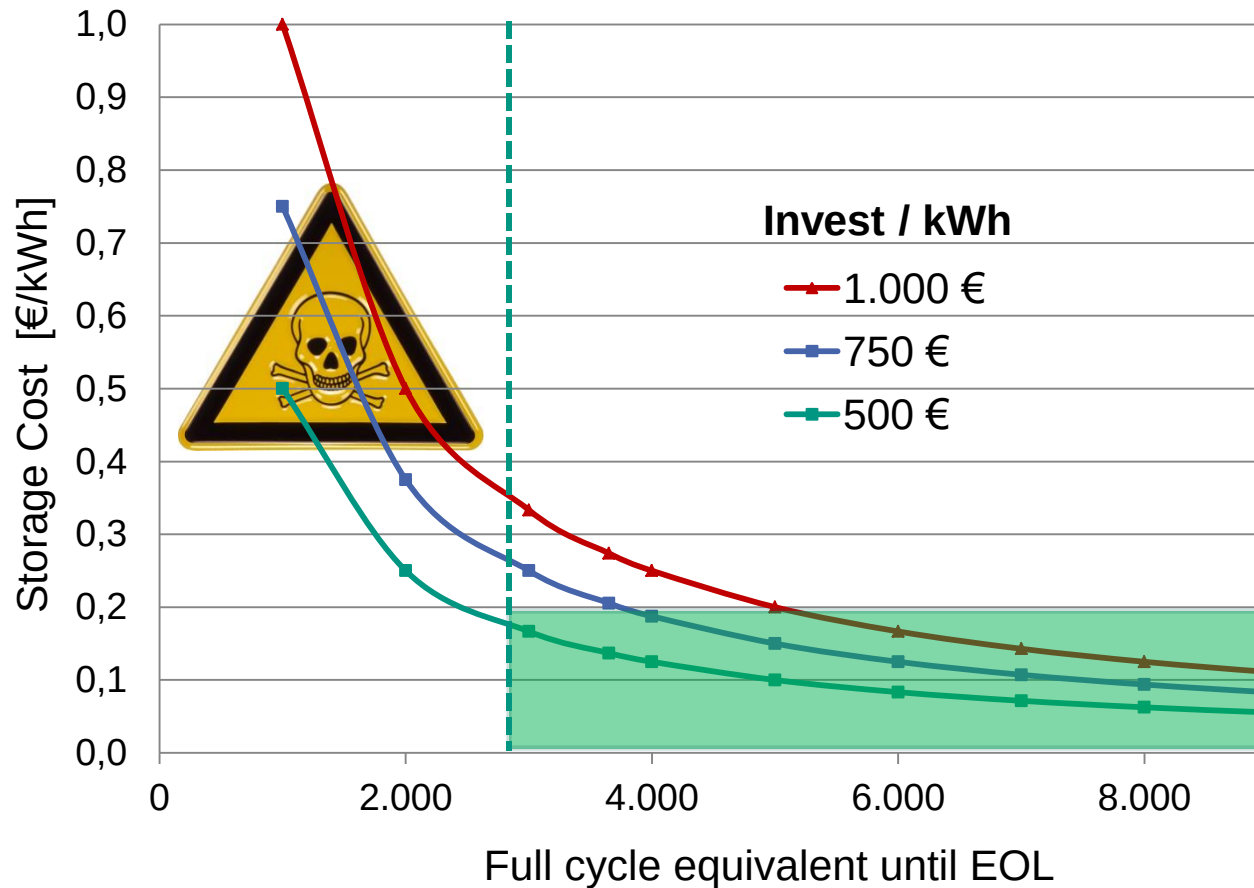
Electricity Costs for PV+Battery Energy Storage Systems (BESS®)

Total Cost of Ownership (TCO)



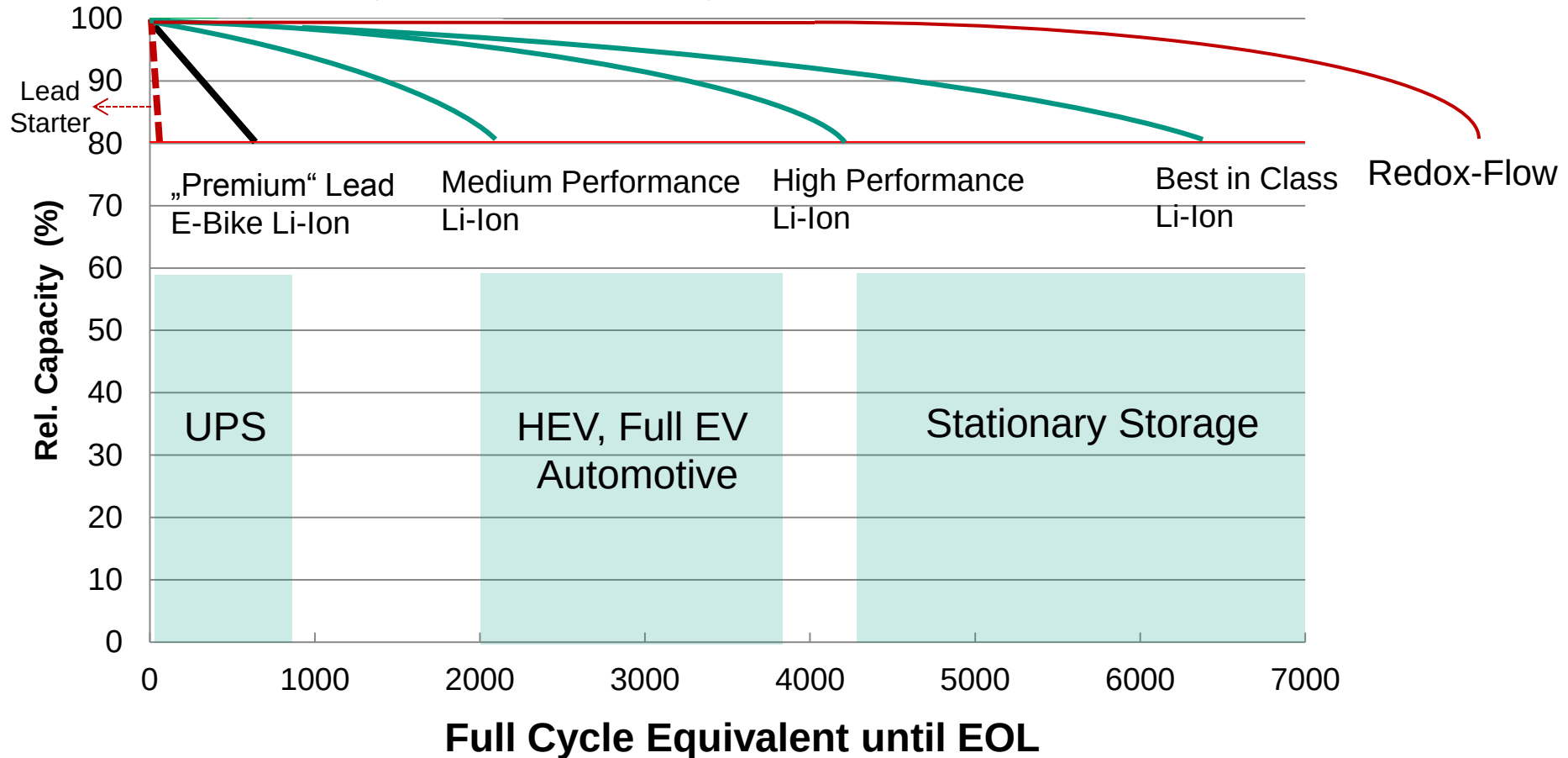
Total Cost of Ownership for Energy Storage

(w/o: PV, interest, maintenance)



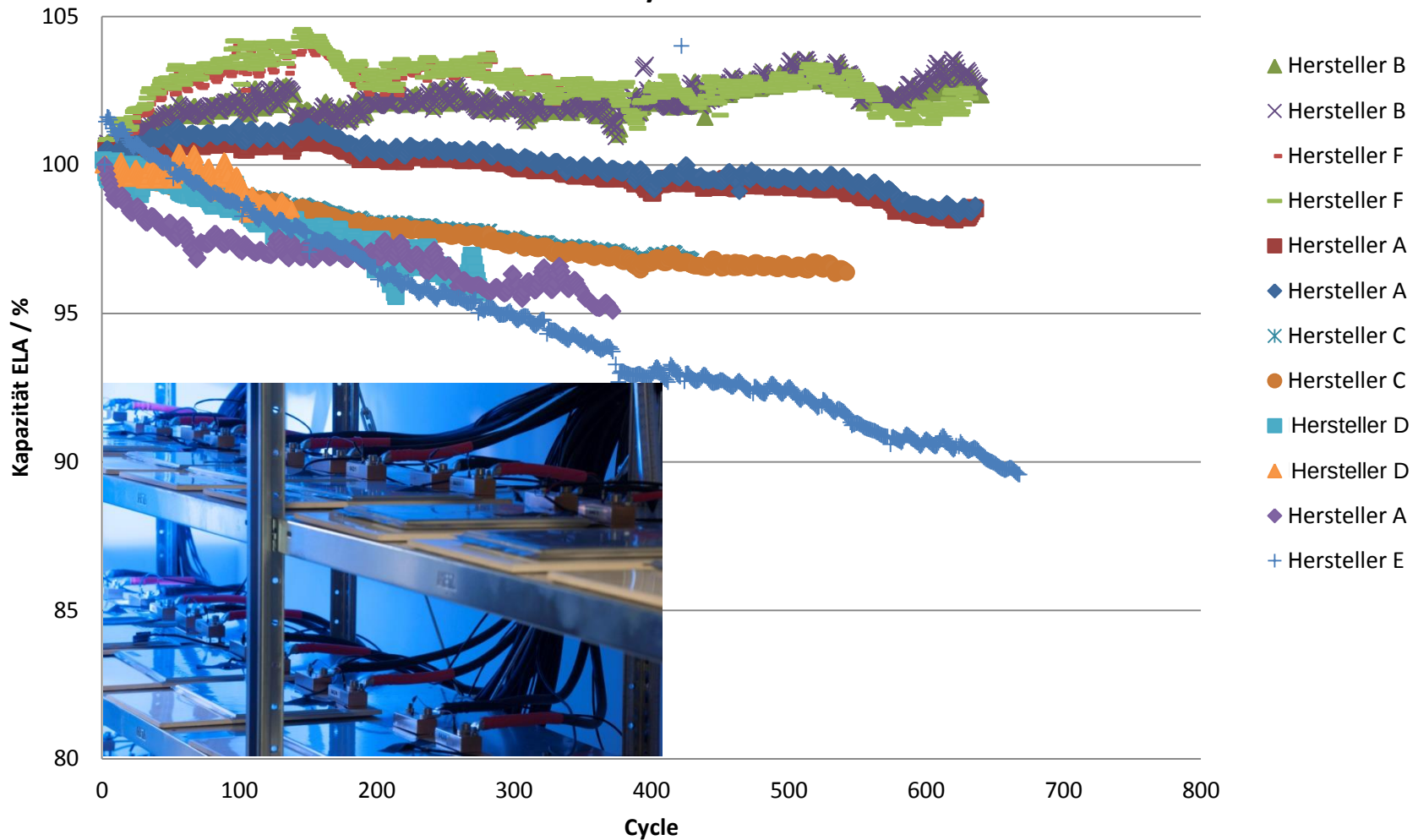
Cycle Life of commercially available Batteries

Results of cycle life tests charge/discharge 1C/1C; 100 % DOD



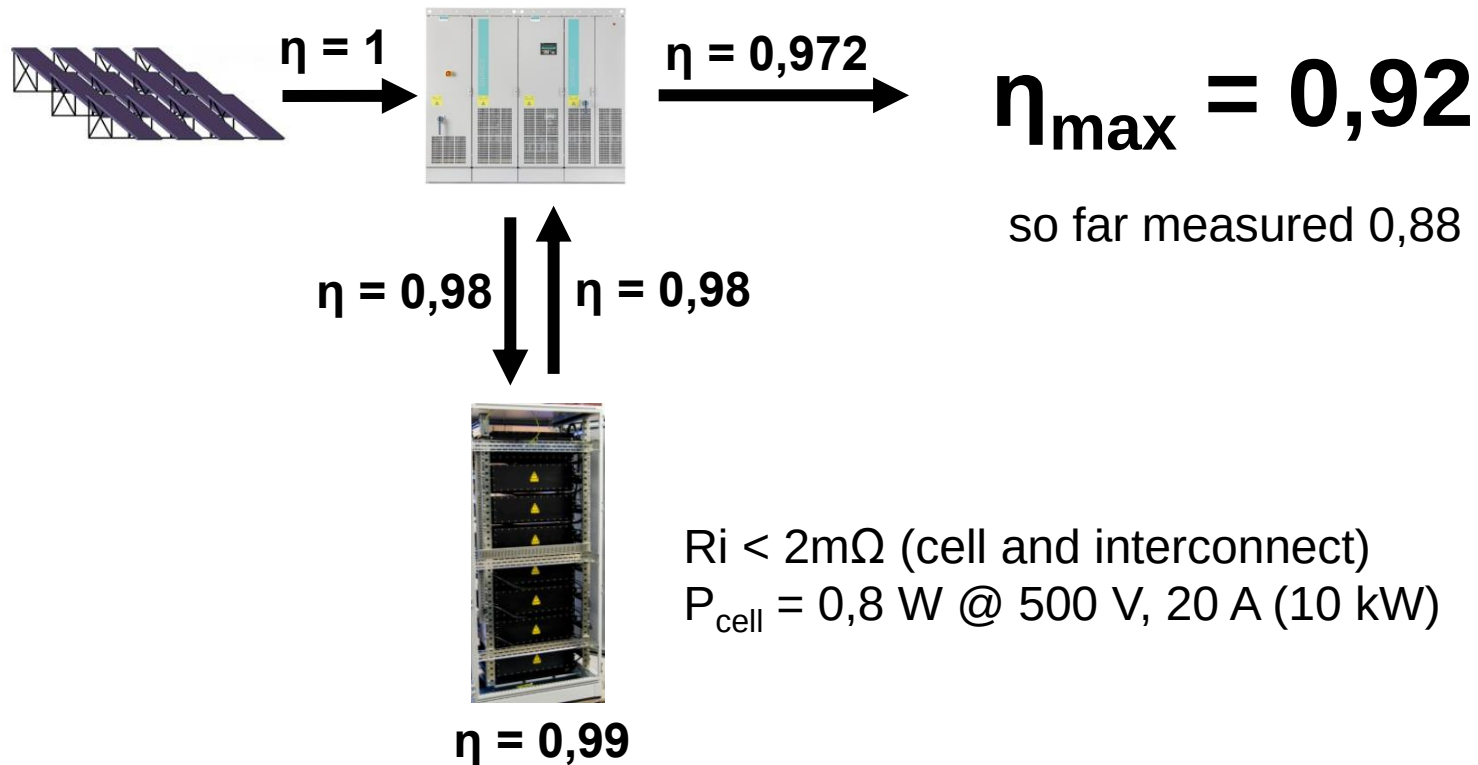
Cycle Life of different Automotive Li-Ion Cells

1C LAD / 1C ELA 100% DOD

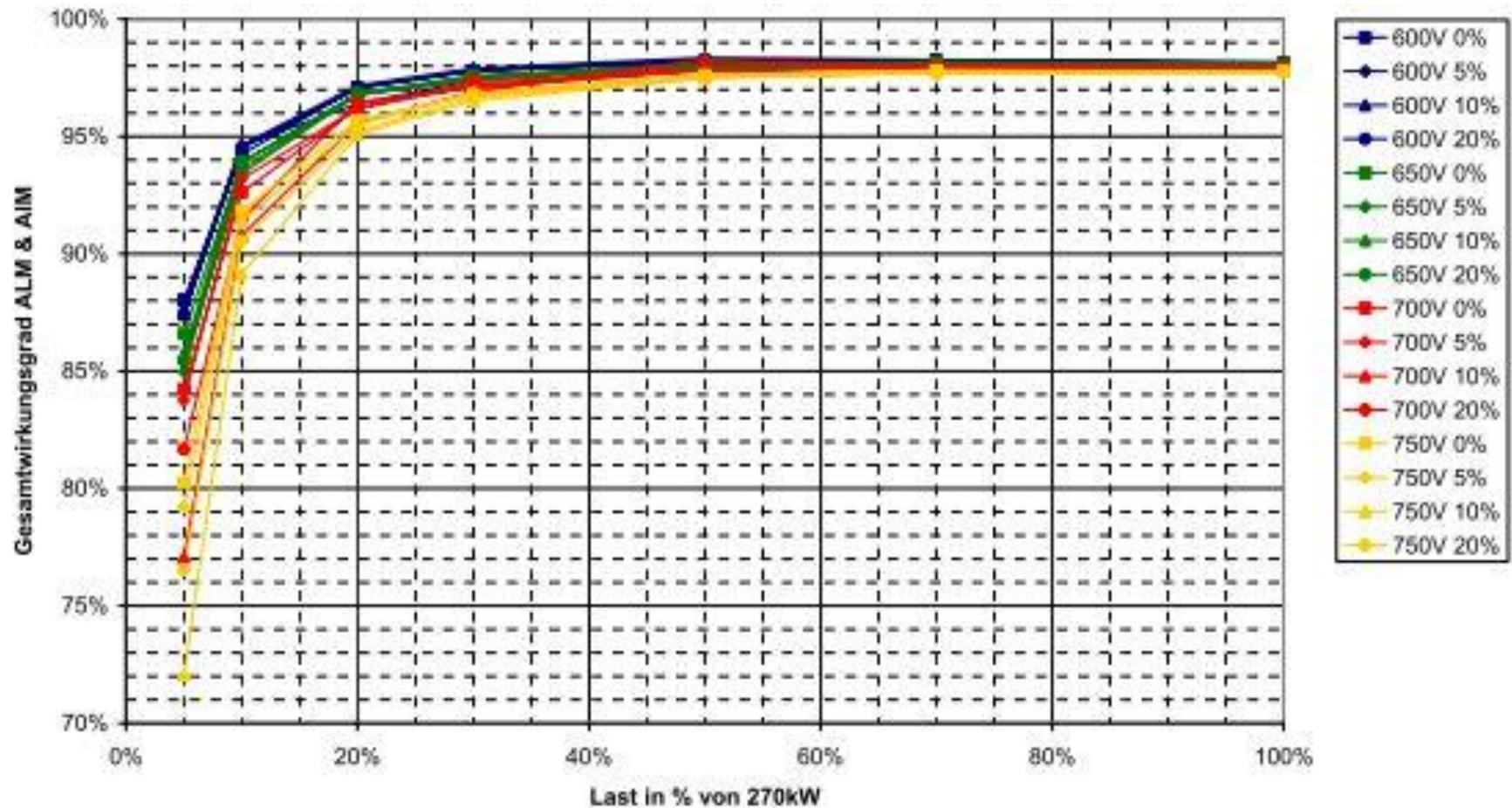


Total System Efficiency

REMINDER: Costs =
$$\frac{\text{energy production costs}}{\text{system efficiency}}$$



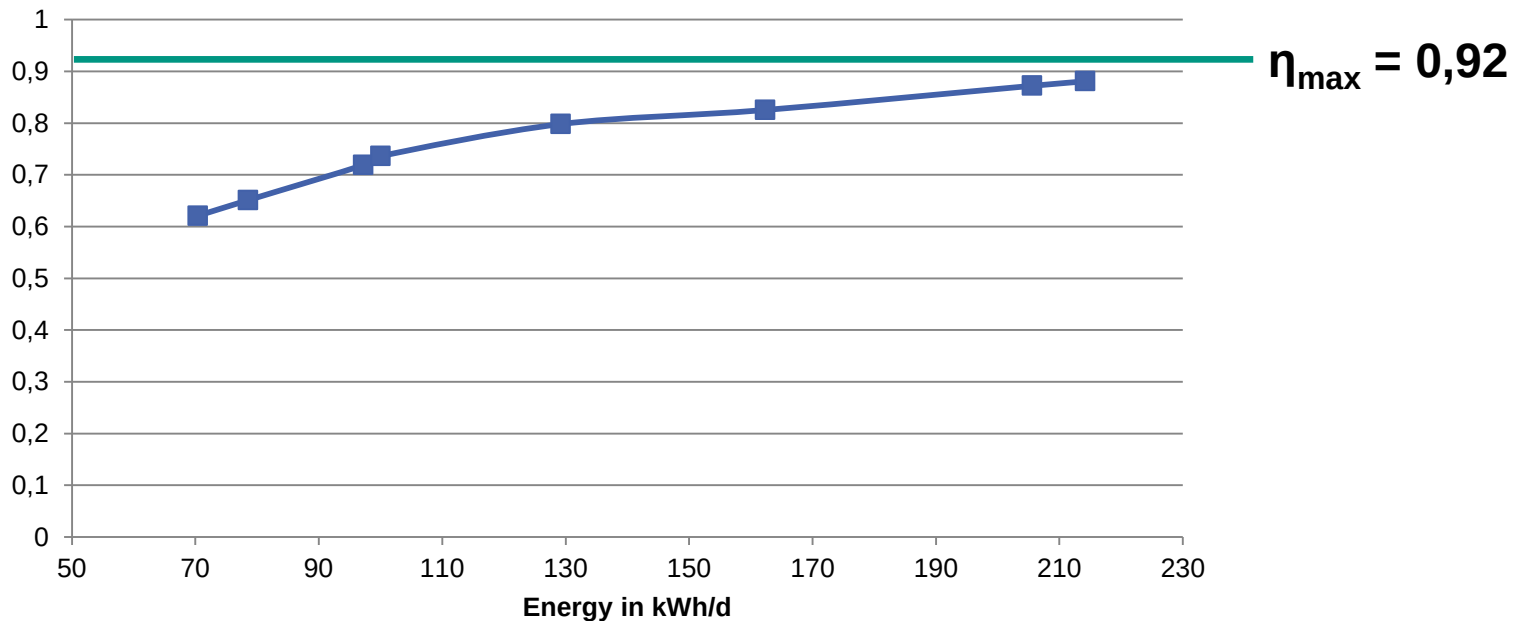
Converter efficiency measured



$\eta = 0,972$ according to EU test procedure (w/o auxiliary systems)

System efficiency measured for 8 days in 2013

Efficiency w/o auxiliary systems



Indication of Electric Energy Costs of BESS®

$$\text{Costs} = \underbrace{\frac{\text{electricity production costs}}{\text{system efficiency}}}_{\text{production term}} + \underbrace{\frac{\text{invest} + \text{maintenance costs} + \text{interest rate}}{\text{full cycle equivalent until end of life}}}_{\text{storage term}}$$

$$\text{Realistic Indication} = \frac{0,07 \text{ €/kWh}}{0,8} + \frac{1400 \text{ €/kWh} + 50 \text{ €/kWh} + 300 \text{ €/kWh}}{6000} = 0,38 \text{ €/kWh}$$

$$\text{Optimistic Indication} = \frac{0,05 \text{ €/kWh}}{0,8} + \frac{1000 \text{ €/kWh} + 50 \text{ €/kWh} + 200 \text{ €/kWh}}{7000} = 0,24 \text{ €/kWh}$$

BESS[®] at KIT March 2013

Jointly developed by:  **AccuSol GmbH** 



Power-Module 250 kW
Battery-Module up to 300 kWh

PV 36 kWp installed
PV 1 MWp in installation

Siemens Power Electronics 270 kW_p



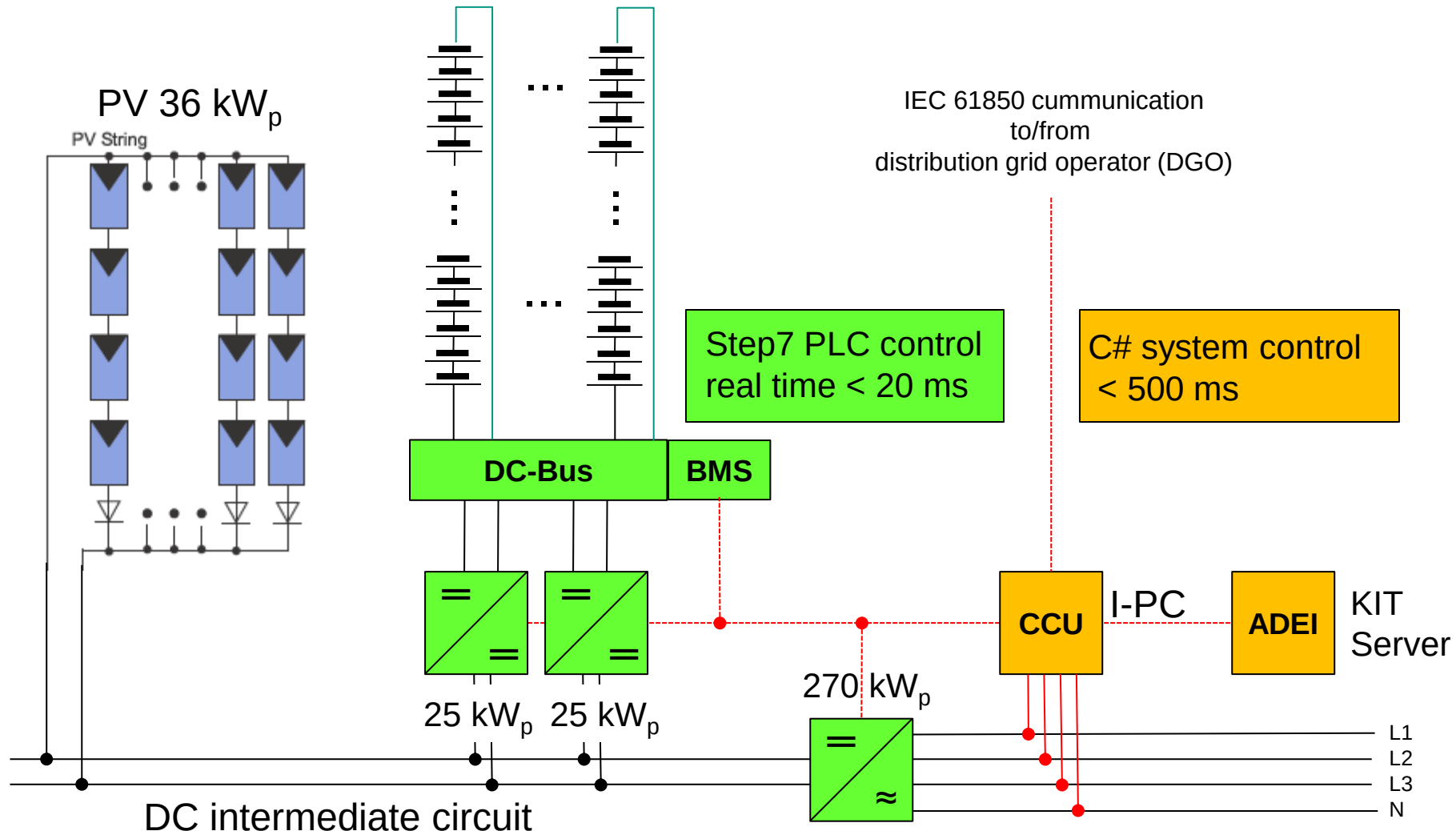
Hardware including frequency, voltage, reactive power / $\cos \varphi$ control

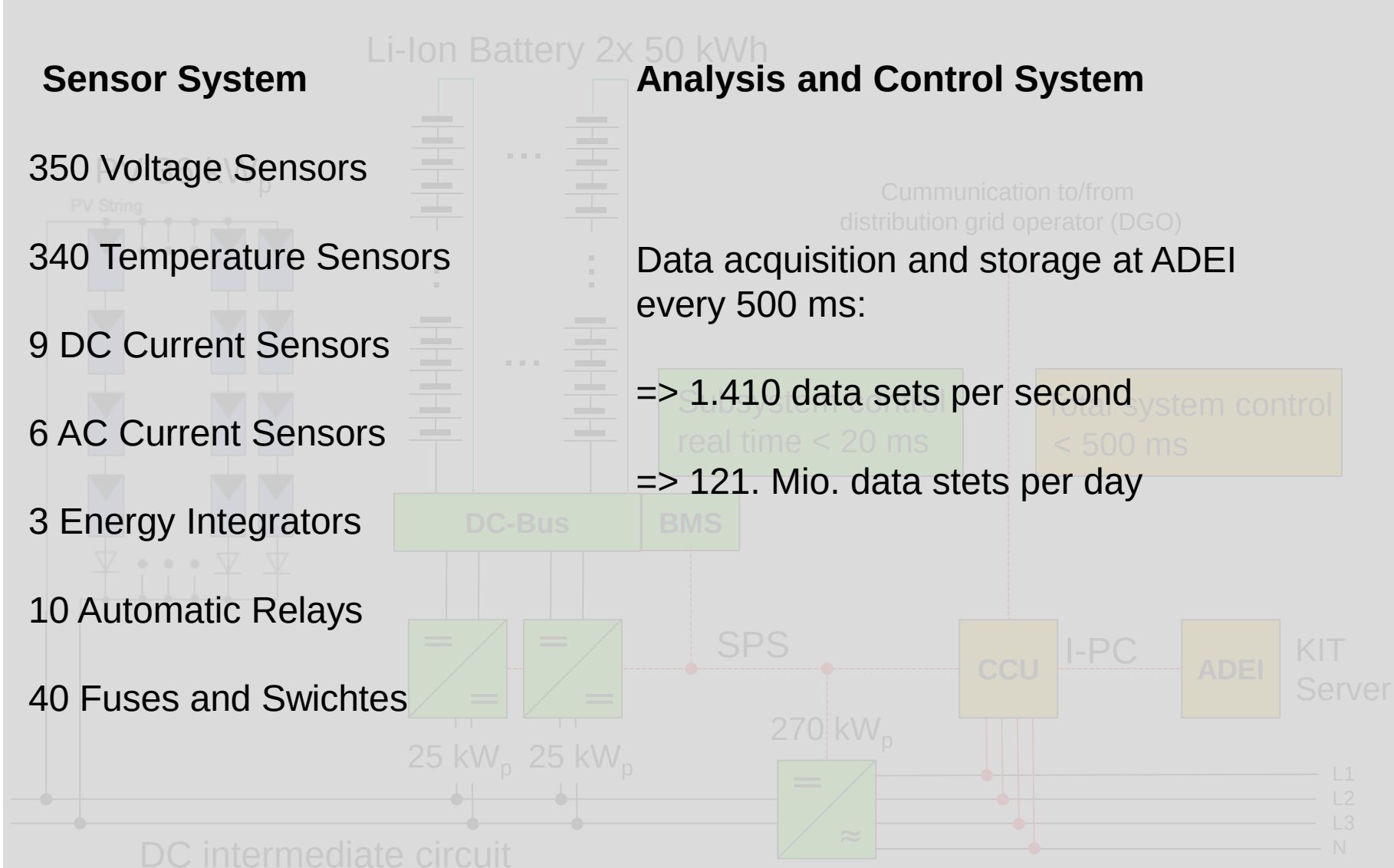
Fully modular Ceramic Li-Ion Technology, 100% made in Germany



Electrical and Control Design of BESS[®] (DC coupling)

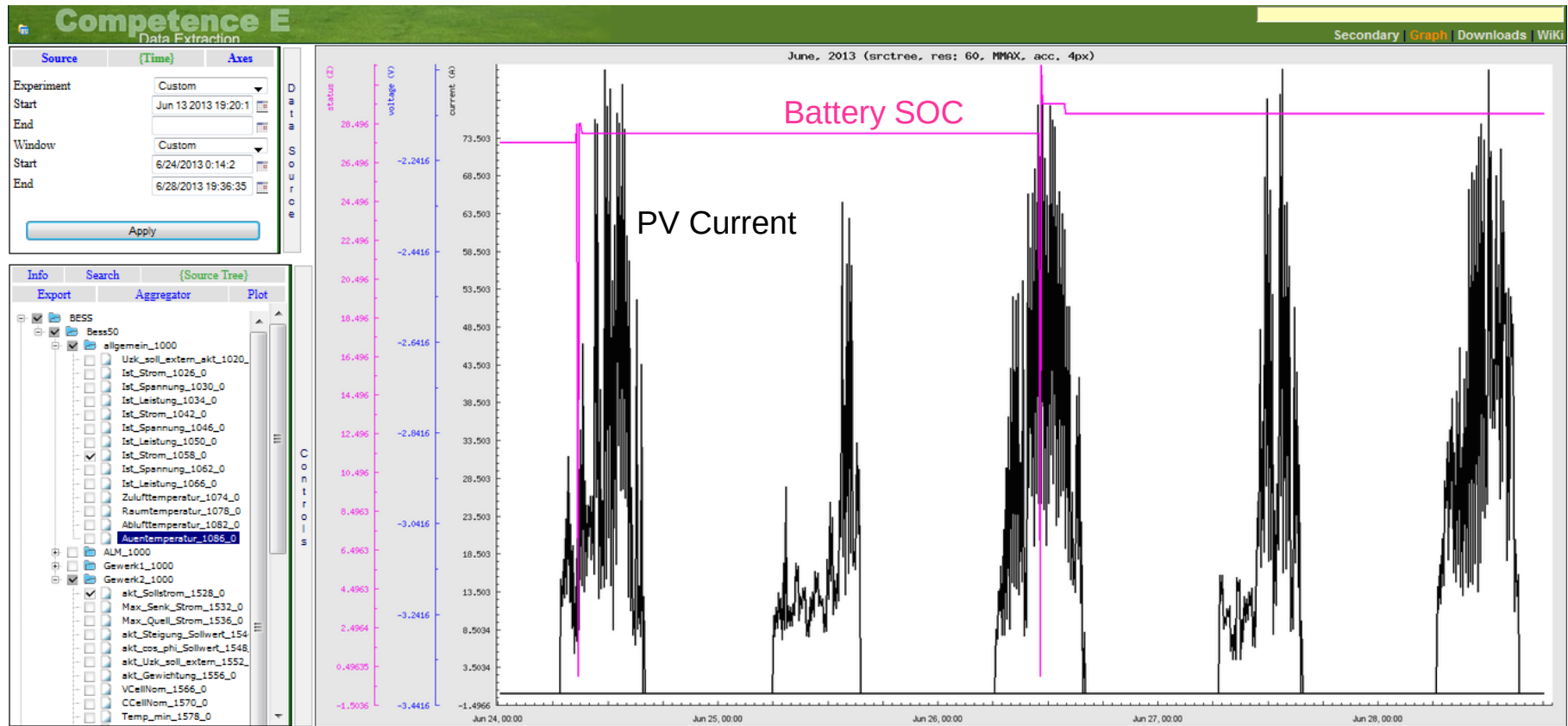
Li-Ion Battery 48 kWh + 24 kWh





Validation of ADEI Data Acquisition

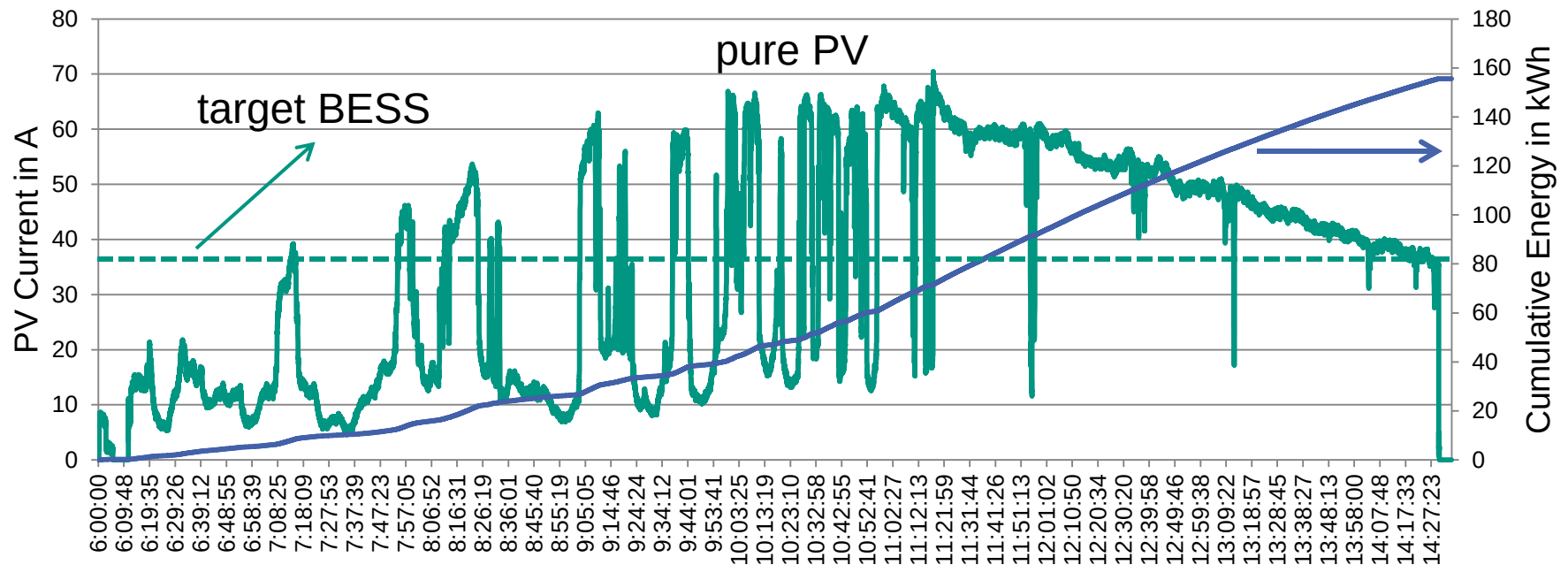
(world wide web access)



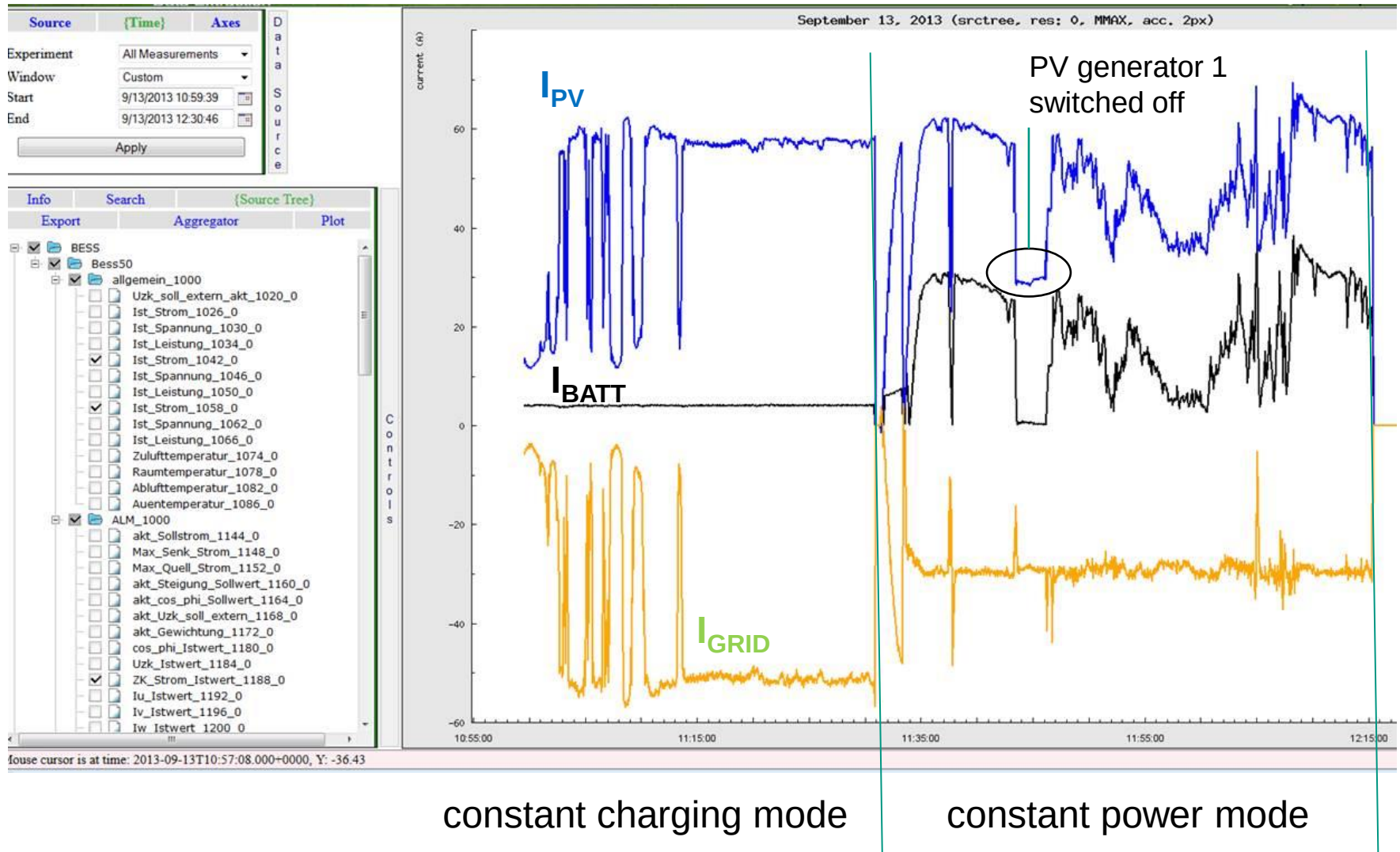
Example: 5 Days in June

Analysis of Fluctuation Frequencies of PV

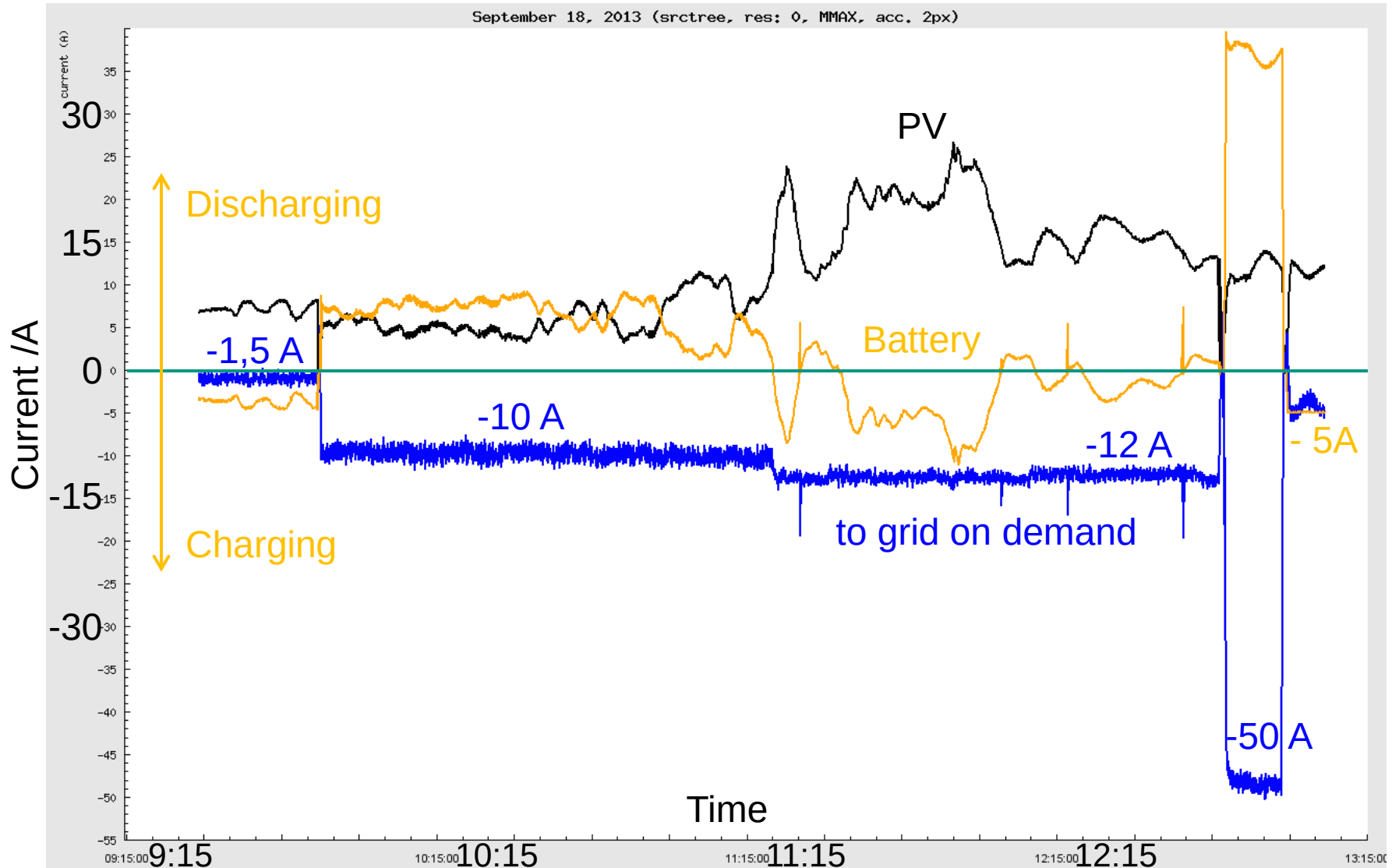
Characteristic frequency is required for optimized control algorithm of BESS[®]



BESS[®] operation in constant power mode

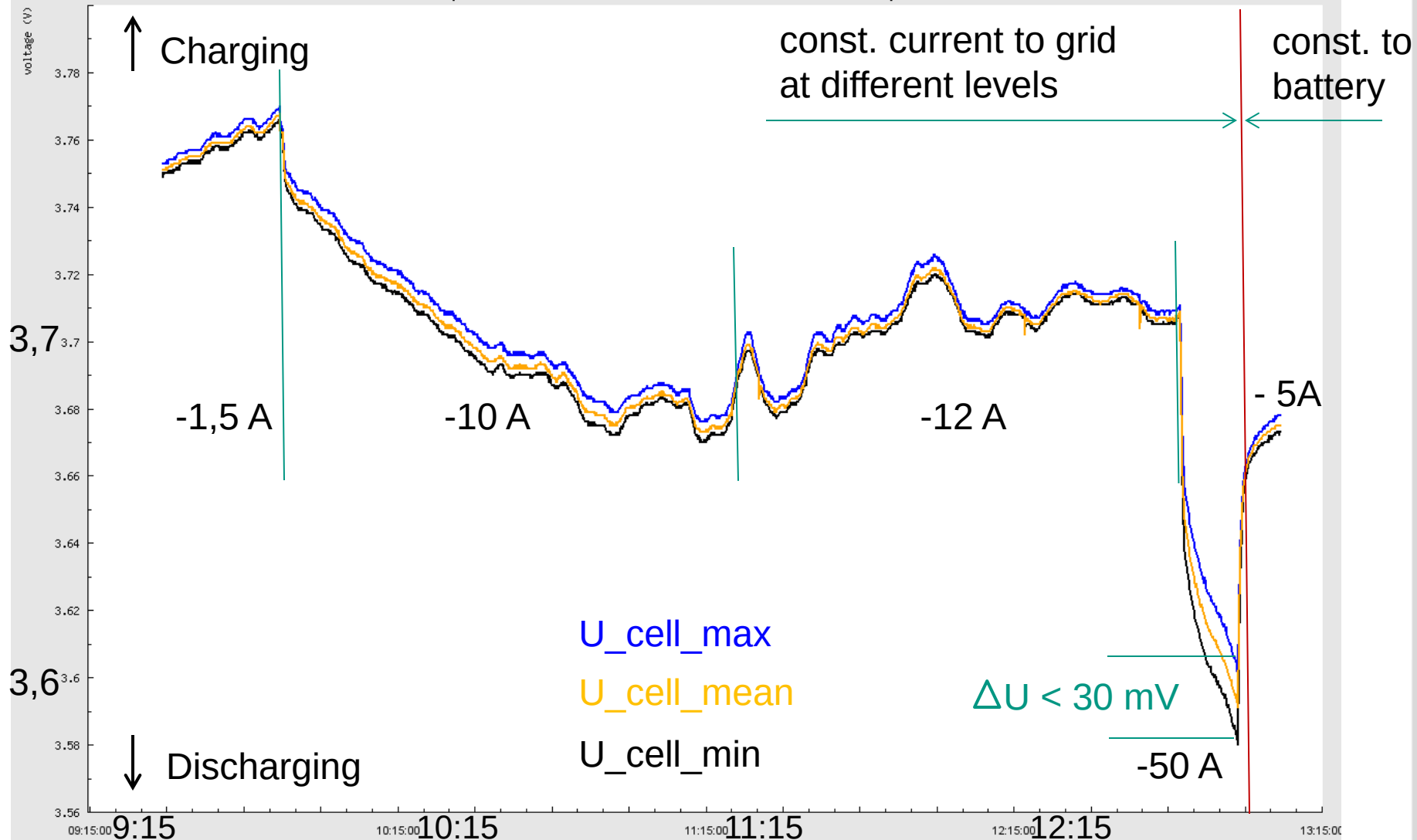


System Currents as of Sept. 18, 2013

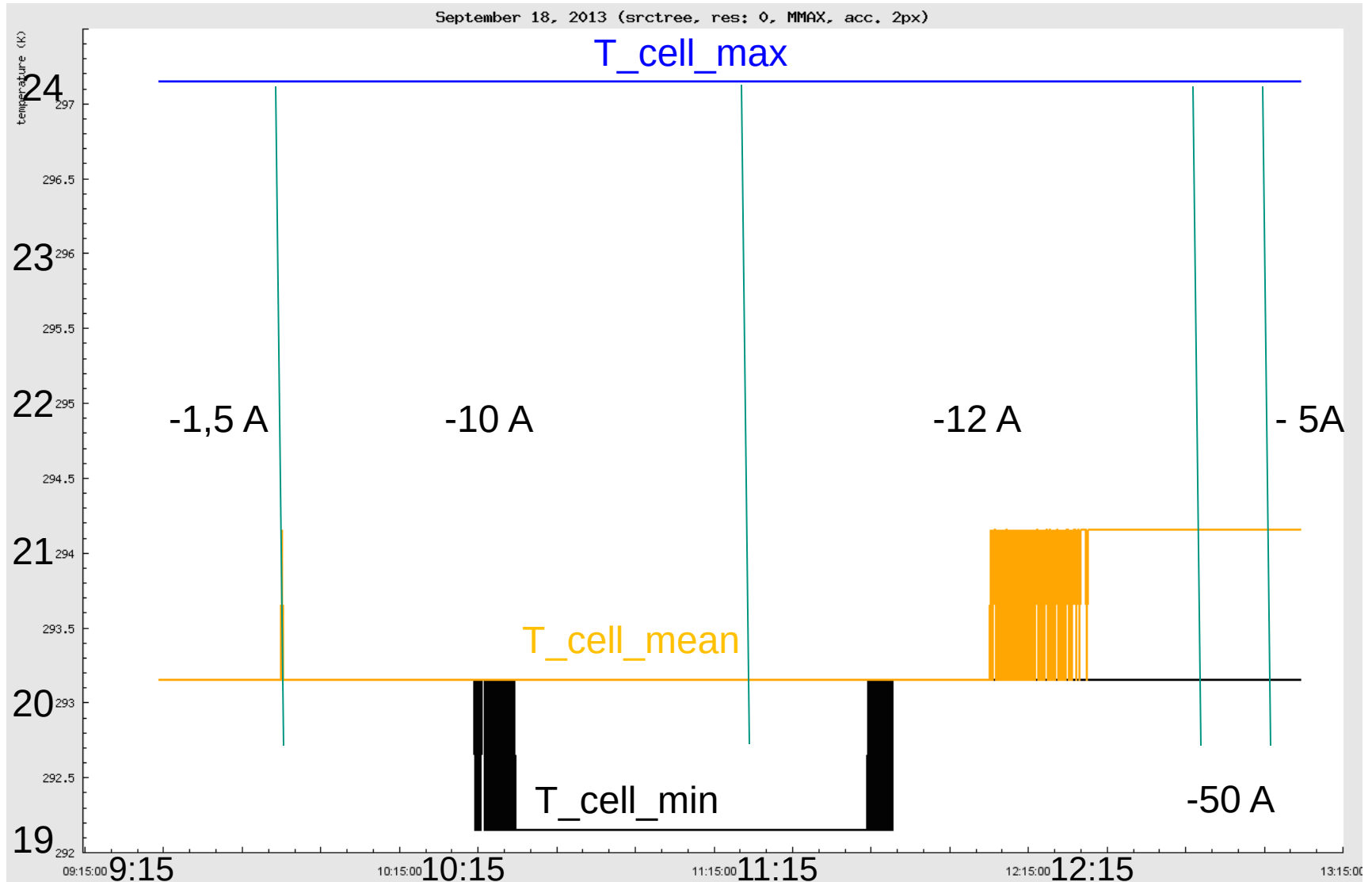


Single Cell Voltages of 168 Cells at Sept. 18, 2013

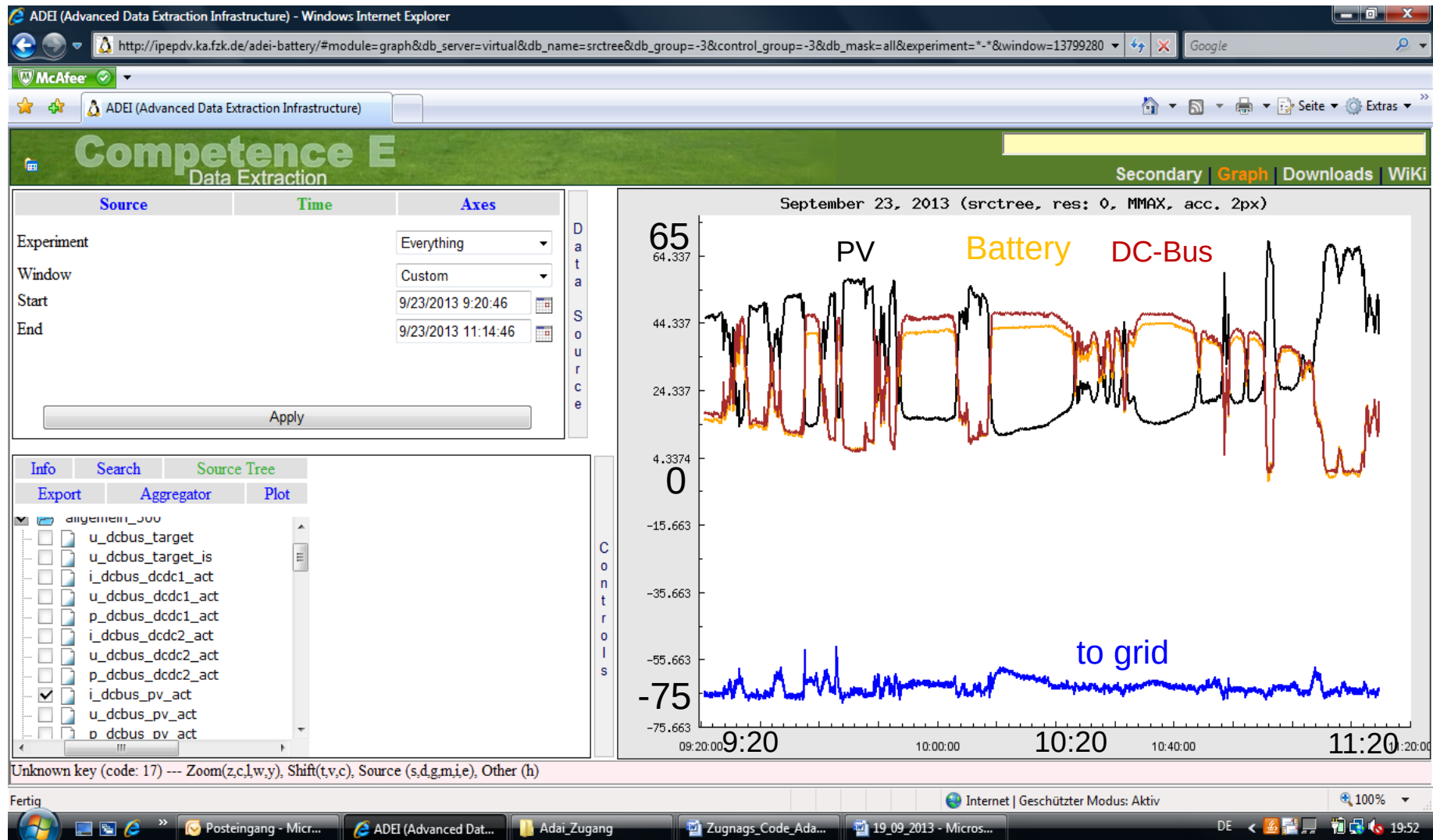
September 18, 2013 (src:tree, res: 0, MMAX, acc. 2px)



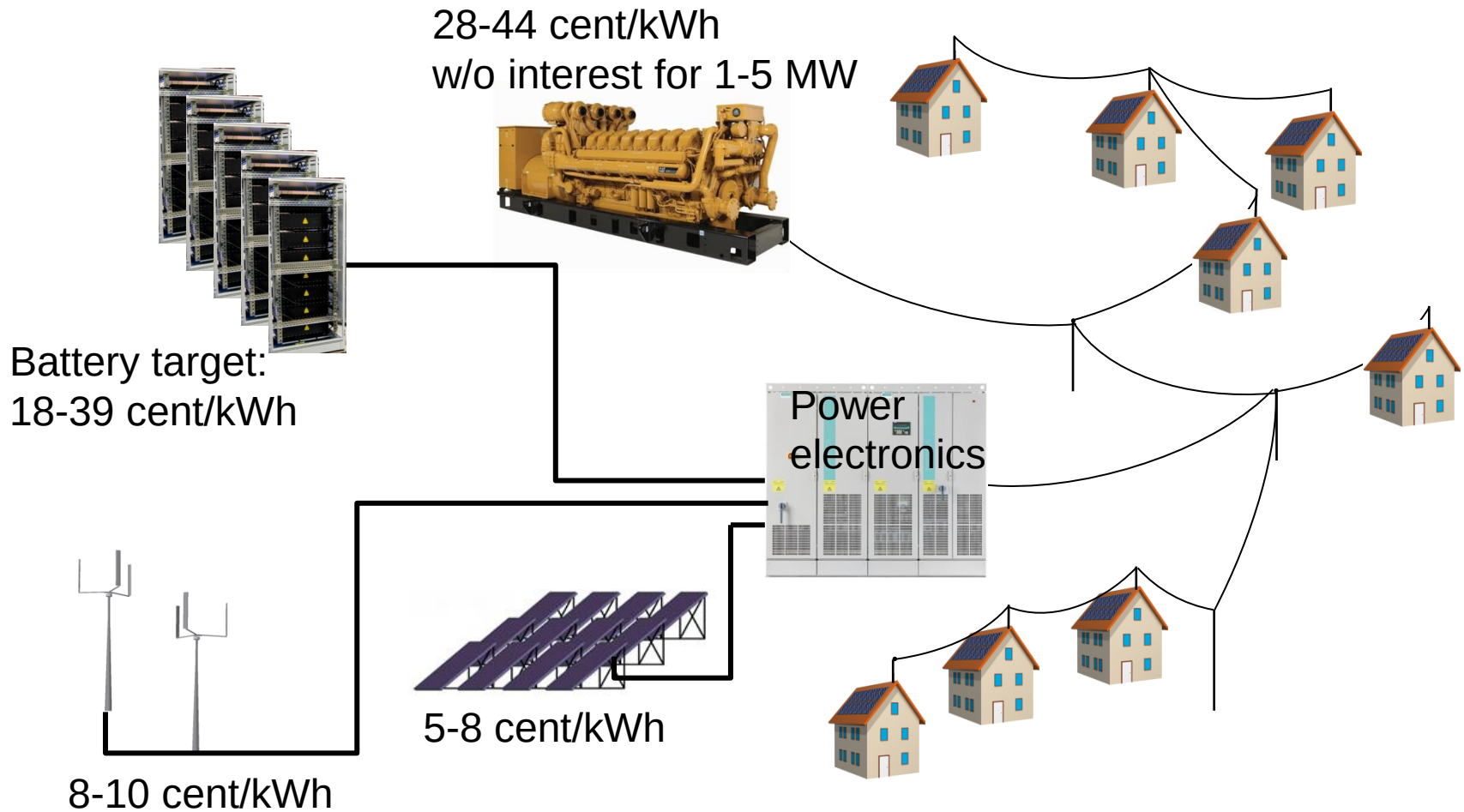
Single Cell Temperatures on Sept. 18, 2013



Searching for the Limits: System Currents



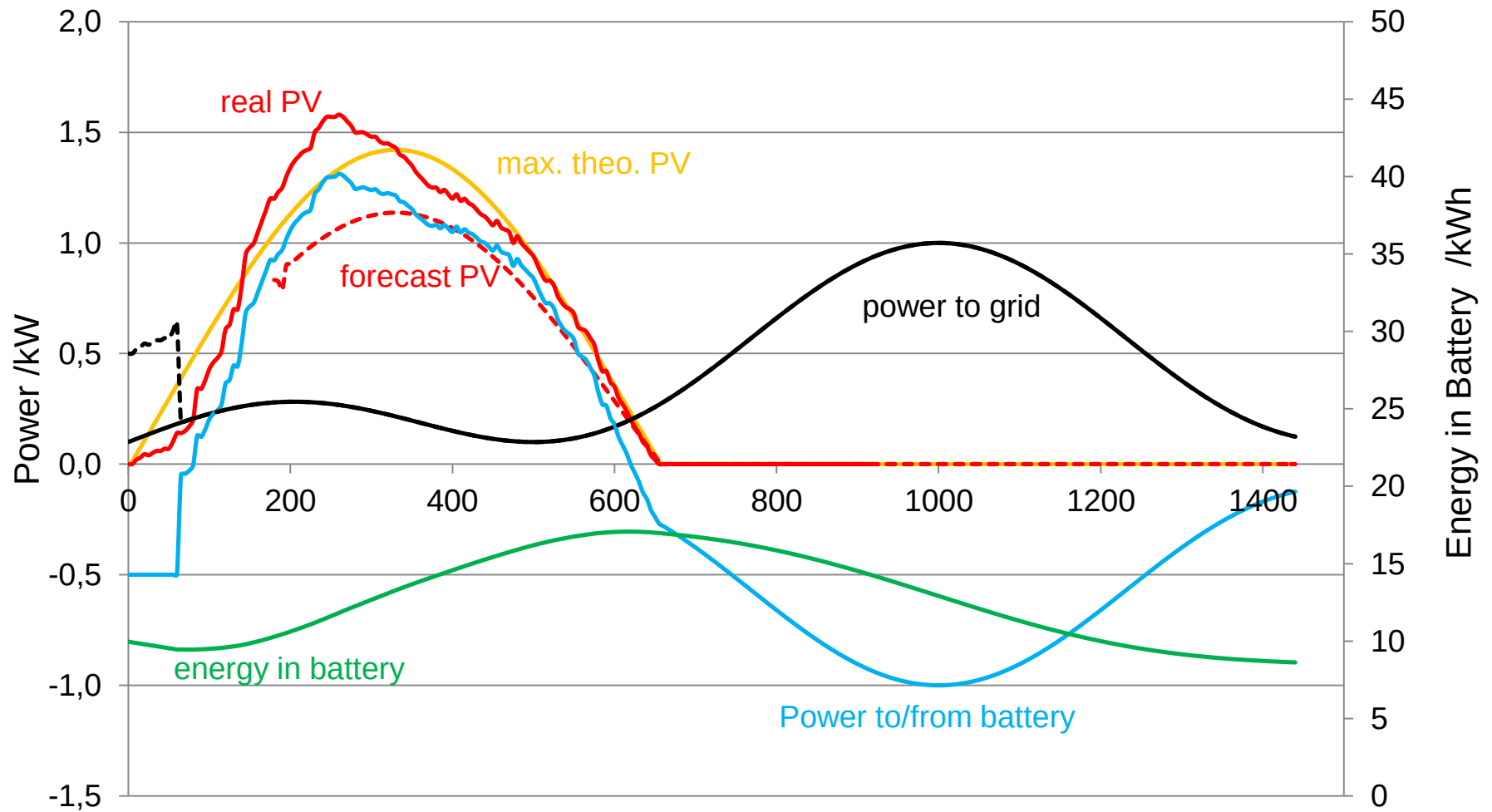
Intregation of BESS[®] in Island Grids



Challenges for the system control of a renewable energy microgrid

- Frequency control
- Voltage control
- Reactive power control / $\cos \varphi$ control
- SoC control and prediction
- Renewables production forecast

BESS[®] predictive control for 04.03.2013



Finally: Safety is not negotiable !!!

Overcharge of a
stationary Li-Ion
battery without
redundant safety
protection

e.g.:

w/o ceramic Li-Ion
cell technology

w/o single cell
temperature
protection

w/o automatic relay

w/o



Our Believe: Safety is not negotiable



End of the overcharge of a stationary Li-Ion battery without redundant safety protection

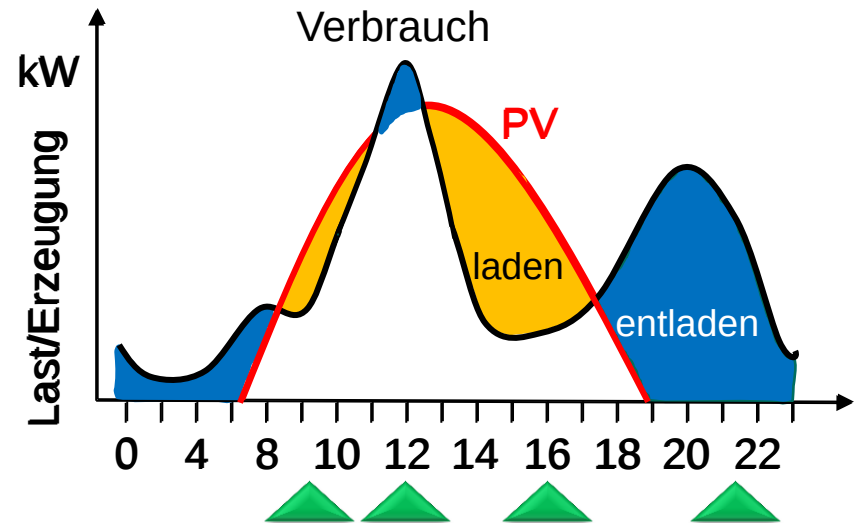
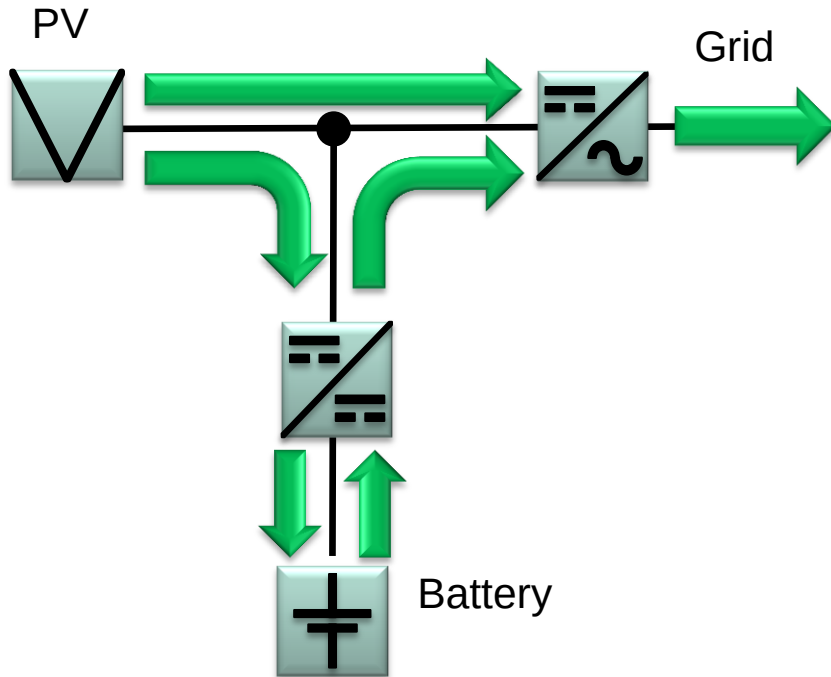
Thank you for your attention!

Competence E



Back-UP

Basic Concept of PV and Battery



Profitability is determined by:

- System design (AC and/or DC coupling)
- System design (power electronics + battery in relation to PV and load)
- System design (control software, reliability, efficiency, maintenance, availability)
- Life time of battery and power electronics